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tRNA database 2026, extended compilation of tRNA sequences and tRNA genes

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Abstract

The tRNAdb database published more than 15 years ago originated as an extension of Mathias Sprinzl's seminal collection of tRNA data dating back to the late 1970s. Throughout its lifespan, the tRNAdb has been used extensively, accumulating more than 1 000 citations. Here, we present tRNAdb 2026, `https://tdb.bioinf.uni-leipzig.de/` **an updated version hosted at Leipzig University. The tRNAdb 2026 features an approximately 10-fold increase in data coverage, containing more than 500 000 new mitochondrial and nuclear tRNA sequences together with the unaltered legacy dataset. Importantly, identifiers and metadata of the legacy dataset remain consistent with the original database, ensuring that previously published references remain valid and findable. The expanded dataset integrates publicly available tRNA annotations and sequences generated using contemporary prediction pipelines followed by semiautomated curation to ensure annotation quality and consistency. In addition, cross-references to external resources such as gtRNAdb facilitate interoperability with existing tRNA annotation frameworks. A modern web-based frontend provides intuitive and efficient access to tRNAdb 2026, while an API enables automated data retrieval and integration into bioinformatics workflows. To support transparent scientific documentation and reproducibility across future updates, we added a versioning system for both current and future database releases.**

Keywords tRNA sequence, tRNA structure, database, tRNA modifications

Abbreviations SeC: Selenocysteine, Sup:Supressor, AC: Anticodon, NCBI: National Center for Biotechnology Information, API: Application Programming Interface, PDB: Protein Data Bank, BV-BRC: Bacterial and Viral Bioinformatics Resource Center, mt-tRNA: mitochondrial tRNA

Key Messages

- The tRNAdb has been greatly expanded
- We provide a modern web based front end for tRNAdb
- We provide an API and release versioning

Introduction

There has been rising interest in tRNAs, promoted by an increasingly broad portfolio of roles beyond their primary function of mediating the genetic code. Some 10 000 publications now link tRNAs with various aspects of cancer, with similar numbers concerning the aspect of evolution of tRNAs and their processing. A solid, well-curated body of data is therefore a key resource for a growing research community. Since the publication of the first "Compilation of tRNA sequences

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and sequences of tRNA genes" in 2005 (19), and its extension as the tRNAdb 2009 (9), a number of databases either dedicated to, or containing tRNA sequences, have been published (3; 1; 18).

When access to the tRNAdb was interrupted due to technical issues in 2024/2025, the data was still widely used. A large number of requests for its continued existence prompted us to re-implemented the tRNAdb on a new, solid technical foundation, and to include an Application Programming Interface (API) for automatic data retrieval. For the present release, we extended its content by leveraging existing nuclear tRNAs from the gtRNA database as well as mitochondrial (mt-tRNA), archaeal and phage derived tRNAs we identified using MITOS2 (7), tRNAscan-SE 2.0.12 (4), and additional specialized approaches. Located in Europe, tRNAdb 2026 contains the most complete collection of mt-tRNA sequences and structures to date.

Implementation

Infrastructure

tRNAdb 2026 was completely reimplemented as a three tier application consisting of a MariaDB relational database, a flask API backend, and a CSS, HTML, and javascript based web frontend. This architecture enables independent change of individual components in order to ensure simple maintenance and future extensions. In contrast to the 2009 implementation (exclusively in Java and HTML/CSS), the new release aligns tRNAdb with modern software design philosophies. Notably, the tRNAdb website utilizes the public API endpoint exclusively for searching and retrieving data. To provide reliable download of large datasets, tRNAdb supports asynchronous data compilation and transfer via dedicated export threads as well as progress reporting of ongoing export jobs via polling.

Data model and provenance

For easy retrieval and global searches, sequences and their most relevant data are stored within a single, fully indexed relation enabling efficient filtering across all sequence categories, attributes, and selected metadata. This includes nucleotide sequence, secondary structure, anticodon, amino acid and origin (used as in the legacy database: genomic DNA-based, mitochondrial DNA-based, or RNA-based sequences including modifications). Every sequence has a taxonomic identifier and a separate table holds relevant taxonomic information obtained from the National Center for Biotechnology Information (NCBI) taxonomy database (17). To facilitate legacy compatibility, the original tRNA dataset (9) remains unaltered regarding sequence, structure, anticodon, amino acid, references, and tRNAdb identifier. We solely updated the taxonomic assignment, i.e., species names and NCBI taxonomy identifiers, to be consistent with the current NCBI taxonomy database. We used the NCBI taxonomy data dump 2026-06-01, accessed 10 June 2026, to create an updated local taxonomic table and reconciled legacy taxonomic assignments using `Bio.entrez` (6) to resolve their current taxonomic information.

The updated relational schema supports a release versioning system which separates the legacy dataset and data contained in tRNAdb 2026 update. Every entry in tRNAdb has a version identifier indicating the first tRNAdb release it appeared in. For every release, data sources (i.e., origin database and version) and software (with corresponding version) used are documented. Additionally, every tRNAdb release is described with a public version string that is displayed on the "Details" page of individual entries, and is also available as an export field. The public version string contains a short description how the sequences published in this particular release were obtained. A detailed description, including software applied and data sources, is available on the "Documentation and Help" page. Sequences in tRNAdb can be filtered by their corresponding release version, allowing users to browse exclusively legacy data, or current releases.

Subsequent releases of the tRNAdb dataset will be published with a corresponding release identifier together with the prior described release metadata. To enable a sustainable, continuous update procedure, we implemented a graphical user interface supported pipeline for import, normalization, and duplicate detection for subsequent tRNAdb versions.

API and programmatic access

With this update we provide a new public HTTP interface, accessible by POST JSON requests at `https://tdb.bioinf.uni-leipzig.de/api`. The API implements OpenAPI 3.1.0 specifications and a thorough documentation for users is available on the "Documentation" page of the tRNAdb website. Fundamentally, our API supports identical search parameters as the tRNAdb website and allows filtering by any combination of origin, species, clade, NCBI taxonomy ID, amino acid, anticodon, sequence and structural motifs, reference (genome accession, external source, publication), and release version.

For many fields, multiple values can be supplied and entries matching at least one are returned. Individual records can be precisely retrieved by using the stable identifier `trnadb_id` which also remained unaltered for the 2009 legacy dataset.

No prior registration, and no personal information (e.g., email address, API key) is required to access the tRNAdb search API at `https://tdb.bioinf.uni-leipzig.de/api` but user requests are limited to one request per second and at most 1000 entries are returned per request to ensure fair use. Every API response, given the request was valid, contains a `total_results` field that states the total number of sequences matching a request and all relevant results can be paginated via an offset parameter. For requests that match a large number of records, and hence constitute bulk retrieval of tRNA sequences, a second API endpoint provides export to TSV, CSV, JSON, or FASTA which is available at `https://tdb.bioinf.uni-leipzig.de/export`. Note that the unique tRNAdb identifier connected to every entry is stable, hence over changing releases, existing identifiers will not change and remain relevant for retrieval purposes. For usage details see the documentation page of the tRNAdb website.

Website

The tRNAdb web interface (see Figure 1) can be found at `https://tdb.bioinf.uni-leipzig.de/`. It provides user-friendly access to the tRNA data stored in our database and can be functionally separated into three subpages: "Search", "Details", and "Taxonomy Tree". The main functionality of the website is the "Search" page which allows users to filter tRNA by organism, origin (genomic DNA-based, mitochondrial DNA-based, or RNA-based including modifications), amino acid, anticodon, membership of taxonomic clades, external references (publications, genome IDs, external datasources) as well as sequence motifs and structural constraints. Species can be filtered precisely via NCBI taxonomy ID as well as their scientific species name and species synonyms as annotated in the NCBI taxonomy

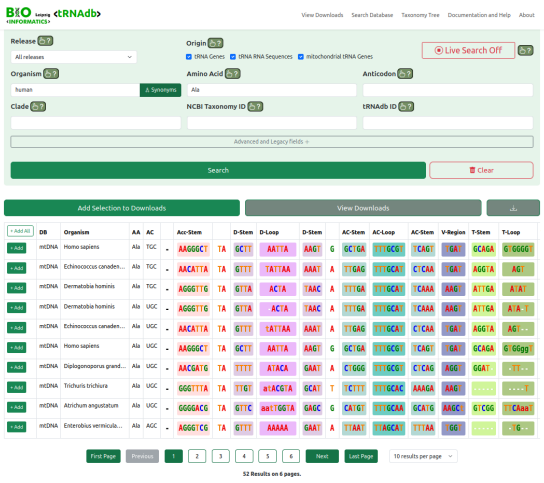


Figure 1 Example of the tRNAdb website, showing the first page of the results for a search for *human* tRNAs for alanine, searching all three origins and using synonyms from NCBI taxonomy.

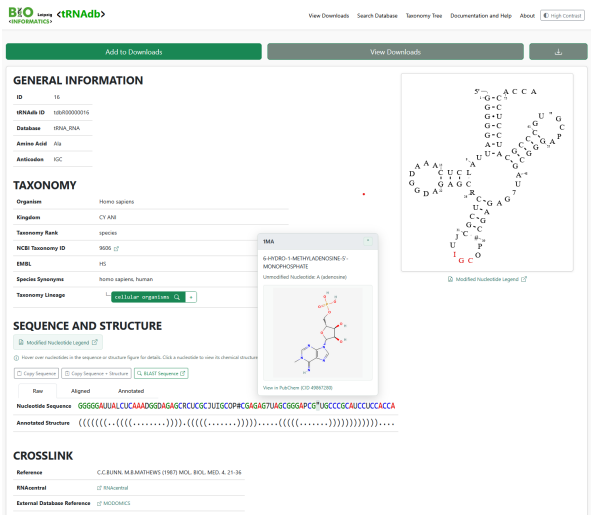


Figure 2 Example for the Details page of a *homo sapiens* tRNA for alanine, including modifications. Mouse-over shows information about the modified nucleotide 6-hydro-1-methyladenosine-5'-monophosphate.

database. For every species with tRNA sequences present in tRNAdb, their taxonomic lineage, as annotated in the NCBI taxonomy database, is stored and tRNA belonging to a specific clade can be obtained by using the "Clade" field of the search page.

Upon a successful search, the user is presented with a paginated list of matching tRNA sequences with a snapshot of data available for every sequence: origin, organism, amino acid, anticodon and its structurally annotated nucleotide sequence (separated into acceptor stem, D-loop, D-stem, etc.). Every result entry, or every entry matching the current search request, can be stored in a temporary selection which can be committed to persistent storage as a named collection for later access or download.

A download manager allows exporting of individual collections, or every saved collection, and offers customizable export options. Export of data in `TSV`, `CSV`, `JSON`, and multi line FASTA format is supported and every export can be further customized by selecting various fields that are relevant to the individual user (e.g. sequence, structure, annotated sequence, species name, taxonomic lineage, reference, and more). FASTA export supports a customizable sequence identifier, for which a preview is generated. For every tRNA sequence entry in our database, a "Details" page is available (see Figure 2). This page contains general information (e.g., unique identifier, origin, amino acid, anticodon), taxonomic information, sequence and structure, crosslinking information together with relevant links to external resources, and additional metadata if available. Furthermore, a standardized visualization of the secondary structure with a highlighted anticodon is displayed for every entry.

Modified bases are displayed using an expanded legacy one letter code. We implemented a mouse-over for the sequences and the structure image which shows the name, unmodified nucleotide of origin, and the PubChem (10) structure of the compound described by the one letter code (see Figure 2).

We implemented a searchable taxonomy browser which displays an expandable tree view of the NCBI taxonomic database. Every node in the taxonomy tree is annotated with its scientific name according to the NCBI taxonomy as well as its unique NCBI taxonomy ID. The taxonomy browser only displays taxonomic groups for which

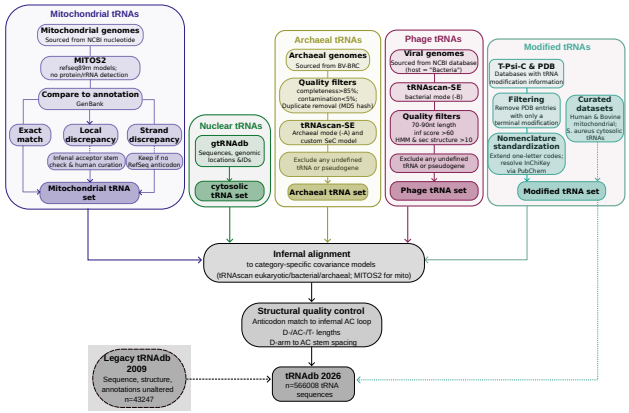


Figure 3 Workflow for data inclusion. Different approaches for data acquisition and pre-filtering for different sources of tRNAs as well as common filtering and alignment steps are shown.

there are tRNA sequences present in tRNAdb. For every entry in the taxonomy browser, the number of tRNA sequences separated by origin is displayed. Every node in the taxonomy browser contains a link, which allows to search tRNAdb for sequences belonging to this taxon.

Inclusion of new data

To include new data into tRNAdb, we used different approaches for different types of tRNA for candidate identification, followed by alignment and quality control for all candidate entries except for the modified mitochondrial RNAs where the published assignment of bases to canonical locations was used (20; 21), see Figure 3.

Mitochondrial tRNAs

To expand the set of mitochondrial tRNAs included in the tRNAdb, a total of 16 630 mitochondrial genome sequences, as well as their annotations were obtained from NCBI Nucleotide (16), accessed October 2024. We used MITOS2 (7) without detection of proteins and rRNA and the refseq89m models on each mitochondrial genome to search for tRNAs. The predicted sequences were then compared to the NCBI annotation. All tRNAs that matched exactly to the NCBI annotation were added to tRNAdb. In case of discrepancies of location, we checked whether the structure of the acceptor stem predicted by Infernal 1.1.4 (13) passed human curation. In the affirmative case, the sequence predicted by MITOS2 was deemed correct and included into the database. In case of discrepancies in the assignment of the strand (reading direction), we retained the MITOS2 prediction if no plausible anticodon could be identified in the anticodon stem and loop of the NCBI annotation. After filtering (see "Alignment and quality control section" below), this resulted in 305 905 novel predicted mt-tRNAs that were added to the `tRNAdb`.

Nuclear tRNAs

Genomic tRNAs were sourced from the GtRNAdb, using the data release from 2024 (3). Information retrieved included sequences, genomic locations and genome IDs. After filtering, 216 856 novel genomic tRNAs were added to the tRNAdb. GtRNAdb contains only a limited number of archaeal tRNA entries (below 9 000), covering a narrow taxonomic range within archaea. Since GtRNAdb also contains no virus-derived tRNAs, we decided to further expand tRNAdb in these two areas.

Archaeal tRNAs

Archaeal genomes were sourced from the Bacterial and Viral Bioinformatics Resource Center (BV-BRC) database (14), accessed October 2024. Only high-quality genomes were retained for further analysis, and genome quality was assessed by CheckM2 ($\geq$85% complete, <5% contaminated) (5). These archaeal genomes cover a broad taxonomic range and include all major archaeal superphyla (Asgard, DPANN, TACK, and Euryarchaeota). We did not rely on the BV-BRC annotation of tRNA genes, as we found it inconsistent with predictions using the most up-to-date version of tRNAscan-SE 2.0.12 (4). Notably, tRNA-Ile2 is often mislabelled as tRNA-Met in BV-BRC, the most frequent mismatch we observed in isotype assignment (34%). This likely reflects the agmatidine wobble-base modification unique to archaea (4), highlighting the necessity to use domain-specific models. Therefore, all genomes were annotated using tRNAscan-SE in archaeal mode. Selenocysteine tRNAs were identified using the SeC-specific covariance model, since it has an atypical secondary structure (4). As independent validation, all tRNAscan predictions were re-scored with Infernal (13) against tRNAinf-arch and tRNAinf-arch-SeC models. After filtering (see "Alignment and quality control section" below), 21 127 tRNA genes from 228 unique archaeal genomes were added, of which 214 were not represented in the 2009 tRNAdb, substantially expanding archaeal coverage relative to the legacy dataset.

Phage tRNAs

Genome sequences of all viruses annotated with the host term "Bacteria" were obtained from NCBI Virus (12), accessed 17 August 2026. Putative tRNA genes were identified using tRNAscan-SE v2.0.12 (4) in bacterial mode. Predicted tRNAs were retained only if they were 70–90nt in length, had an overall tRNAscan-SE Bit Score >60, and had both HMM and secondary-structure scores >10. Predictions with undefined isotypes or classified as pseudogenes were excluded. After filtering (see "Alignment and quality control section" below), 7 099 viral tRNA genes from 1 558 unique genomes were added to tRNAdb.

tRNAs sequences including modifications

Modified tRNA sequences, as well as their genomic variants, were obtained from the T-Psi-C (15) database, the MODOMICS (18) database, and the Protein Data Bank (PDB) (2). Sequences retrieved from the PDB having only a single modification located at the first or last base were not included into the database. In order to display alignments properly, legacy one-letter codes for modifications were retained. The dictionary of one-letter codes for modifications was extended to include any that were not present in the legacy tRNAdb 2009. For these novel modifications, we also identified the compound name using the PDB entry. The InChiKeys and InChis were extracted from PubChem (10) to check whether the compound was already present in the MODOMICS database. A list of all modifications with the one-letter codes used in tRNAdb and MODOMICS can be found in the supplement and on the database webpage. Mature mitochondrial tRNA sequences from *H. sapiens* (21) and *B. taurus* (20) and cytosolic tRNA sequences from *S. aureus* (8) were added from the supplement of the respective publications. The included canonical base numberings were used directly for alignment and structural annotation.

Alignment, quality control, and secondary structure

For all novel sequences, we used Infernal (13) to align them to the respective tRNA covariance models (eukaryotic, bacterial or archaeal models from tRNAscan for nuclear tRNAs, the respective MITOS2 models for mitochondrial tRNAs). When an anticodon was annotated, we compared it to the one identified by Infernal (at the center of the anticodon loop). To filter tRNAs with suspicious structural features, we also checked the length of the D-loops, AC-loops, and T-loops, as well as the number of unpaired bases between D-stem and AC-stem. For the AC-loop, we used a strict length of 7 nt. For the other structural features, we used the maximum length of the respective structural feature in tRNAdb 2009, allowed a single nucleotide difference in length (resulting in 18 and 28 nt for D and T-loops respectively and 5 nt between D and AC-stem) and discarded a total of 404 tRNAs that did not make one of the four thresholds. As a last filtering step, all entries that were duplicates with respect to sequence and taxonomic ID were unified when necessary, so that every sequence/species pair in the tRNAdb is unique. To provide a measure for the relative stability of the canonical structure compared to other structures the sequence might adopt, we used the ViennaRNA package (11) to compute the frequency of the canonical (unmodified) structure in the ensemble as well as the ensemble defect, a measure of how different the structures of the thermodynamical ensemble are to the canonical structure.

Statistics

The database lists 336 430 mitochondrial tRNA entries comprising 223 895 unique sequences. The vast majority of these originate in

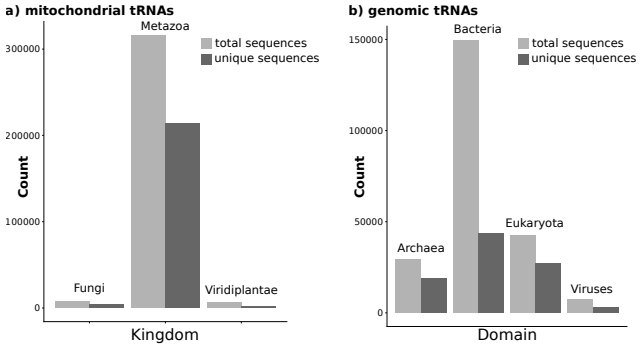


Figure 4 Number of tRNA sequences per clade. a: mitochondrial tRNAs at kingdom level. b: genomic tRNAs at domain level. Light gray shows number of all sequences, dark gray number of unique sequences.

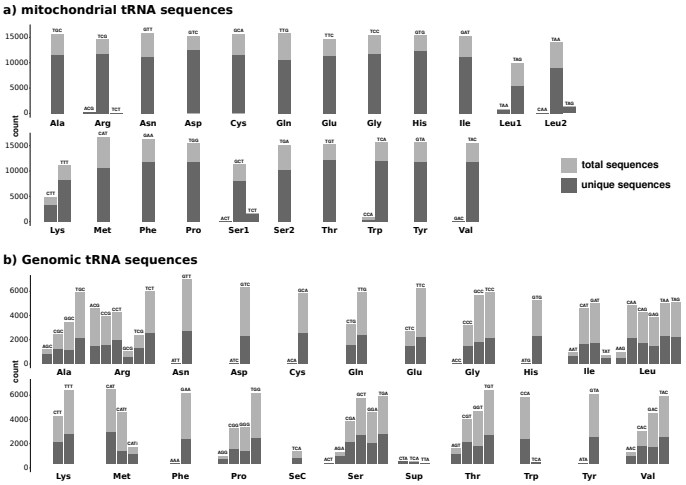


Figure 5 Distribution of tRNAs per amino acid and anticodon. a) mitochondrial tRNA sequences b) genomic tRNA sequences. Light gray shows number of all sequences, dark gray number of unique sequences.

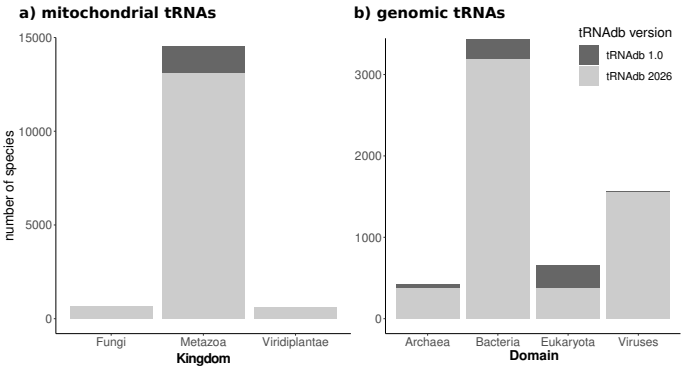


Figure 6 Distribution of species included in the tRNAdb, seperated by clade. a) mitochondrial b) genomic tRNAs. Light gray shows number species in tRNAdb 2026 only, dark gray number of species in tRNAdb 2009.

metazoa (see Fig. 4a), which reflects the distribution of mitochondrial genomes at the NCBI. The genomic part of the database contains a total of 228 955 entries comprising 93 257 unique sequences. The majority, precisely 149 749 entries, are bacterial tRNAs. However, the difference of genomic tRNA representation in tRNAdb is much smaller for unique sequences, see Fig. 4b.

Mitochondrial tRNAs show a relatively even distribution with respect to their abundance, ranging from 14 692 entries for Glu to 16 734 entries for Met. The relatively even distribution can also be seen in the unique sequences: Asp has the most with 12 446, while Ser2 has the least amount of unique sequences with 10 184. The exception to these even distributions are Leu1 (10 857 entries, 6 209 unique sequences) and to a smaller extent Ser1 (12 979 and 9 457, resp.), which code for the two amino acids that have two tRNAs in many mitochondria. Mitochondrial tRNA-Lys is the only tRNA with a notable amount of two different anticodons (Fig. 5a).

In contrast, genomic tRNAs show a much wider variety in their abundance. Leu, being the most abundant, has 23 995 entries and 10 233 unique sequences. Besides the non-standard Suppressor and SeC tRNAs, His has the fewest entries with 5 290 (2 305 unique sequences), while Phe and Trp have the fewest unique sequences with 2 093 (in 5 855 and 5 579 entries, respectively). Notably, many genomic tRNAs (Asn, Asp, Cys, His, Phe, Trp, Tyr) show a clear preference for one codon (see Fig. 5b). Not surprisingly, the number of entries as well as the number of unique sequences per amino acid strongly correlates with the number of frequent (n> 100) anticodons in use for an amino acid ($\rho > 0.93, p < 10^{-8}$ for both).

In comparison to the first version of tRNAdb, we greatly expanded sequences belonging to clades that were either not included before, such as mitchondrial tRNAs from viridiplantae or fungi, or had only very sparse entries, such as archaea and viroidal genomic tRNAs (see Fig. 6). In total, tRNAdb 2026 contains genomic entries from 6 086 species, and mitochondrial entries from 15 796 different species.

Conclusion

The updated tRNAdb has been expanded 11-fold and its infrastructure was modernized throughout. Alongside a redesigned web interface, it now provides a public API and a considerable increase in the number of included entries. Among these new entries now are modified mitochondrial tRNAs and, for the first time, mitochondrial tRNAs from plants and fungi. tRNAdb thus continues to serve as an important resource for the tRNA community.

However, because most entries in the tRNAdb stem from semi-automatic annotation, fringe cases that deviate from established tRNA properties are likely to be rejected, even though these are often of high biological interest. We therefore rely on the community to report such sequences, so they can be incorporated into future releases. This feedback may in turn motivate updated search algorithms capable of capturing a broader range of non-standard tRNAs. To address such cases, we provide a web interface where experts can submit entries of non-standard tRNAs that we will curate and include into the database.

Conflicts of interest

C.L.B. is a co-founder and scientific advisor to Locus Biosciences and a scientific advisor to Quidditas Therapeutics. The other authors declare that they have no competing interests.

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Data availability

The data underlying the tRNAdb 2026 are available in public repositories like GtRNAdb, NCBI nucleotide and BV-BRC. Links to the sources as well as all other relevant information are provided at the detail pages of the webpage `https://tdb.bioinf.uni-leipzig.de/`.

Author contributions statement

B.J.S.: Software, Data curation, Writing – original draft, Writing – review & editing. **S.H.B.:** Conceptualization, Methodology, Investigation, Data curation, Software, Writing – original draft, Writing – review & editing. **P.F.S.:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Funding acquisition. **E.C.:** Investigation, Writing – original draft, Writing – review & editing. **D.I.V** Validation, Writing – review & editing. **M.F.:** Investigation, Writing – review & editing. **T.S.:** Investigation, Data curation, Writing – review & editing. **F.J.:** Resources, Data curation, Writing – review & editing. **M.M.:** Resources, Data curation, Writing – review & editing. **J.P.:** Resources, Data curation, Writing – review & editing. **C.L.B.:** Supervision, Writing – review & editing.

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